Practice workbook · Finding unknown constants from binomial coefficients

Writing coefficients of a binomial expansion in terms of unknown constants (such as n and a) and using given information to solve for them.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-22

1. Question 5 · part b · 6 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part b only.

5 (a) Expand $(1 + ax)^n$ in ascending powers of x up to and including the term in x^3

Express each coefficient of x in terms of a and n where a and n are constants and $n > 2$

(2)

The coefficient of x is 15 and the coefficient of x^2 is equal to the coefficient of x^3

(b) Find the value of a and the value of n.

(6)

(c) Find the coefficient of x^3

(2)

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1. Question 5 · part b · 6 marks · Working and answer

Working and answer

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2. Question 2 · part a · 6 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

2 Given that

$$1 - \frac{1}{3}x + \frac{5}{36}x^2 + \ldots$$

is the binomial expansion, in ascending powers of x, of $(1 + Ax)^n$

where A and n are rational numbers,

(a) find the value of A and the value of n

(6)

(b) Hence find the value of the coefficient of x^3

Give your answer in the form $-\frac{p}{q}$ where p is a prime number and q is an integer.

(2)

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2. Question 2 · part a · 6 marks · Working and answer

Working and answer

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3. Question 2 · part b · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

2 Given that

$$1 - \frac{1}{3}x + \frac{5}{36}x^2 + \ldots$$

is the binomial expansion, in ascending powers of x, of $(1 + Ax)^n$

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3. Question 2 · part b · 2 marks

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Further practice · Attempt part b only.

Question 2 continued

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(Total for Question 2 is 8 marks)

3. Question 2 · part b · 2 marks · Working and answer

Working and answer

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4. Question 5 · part a · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part a only.

5 (a) Expand $(1 + ax)^n$ in ascending powers of x up to and including the term in x^3

Express each coefficient of x in terms of a and n where a and n are constants and $n > 2$

(2)

The coefficient of x is 15 and the coefficient of x^2 is equal to the coefficient of x^3

(b) Find the value of a and the value of n.

(6)

(c) Find the coefficient of x^3

(2)

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4. Question 5 · part a · 2 marks · Working and answer

Working and answer

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5. Question 5 · part c · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

5 (a) Expand $(1 + ax)^n$ in ascending powers of x up to and including the term in x^3

Express each coefficient of x in terms of a and n where a and n are constants and $n > 2$

(2)

The coefficient of x is 15 and the coefficient of x^2 is equal to the coefficient of x^3

(b) Find the value of a and the value of n.

(6)

(c) Find the coefficient of x^3

(2)

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5. Question 5 · part c · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 10 marks)

5. Question 5 · part c · 2 marks · Working and answer

Working and answer

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6. Question 5 · part a · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

5

$$f(x) = (1 + ax)^n \text{ where } n > 2$$

(a) Write down the expansion of $f(x)$ in ascending powers of x up to and including the term in x^3
Give each coefficient in terms of n and a

(2)

In the expansion of $f(x)$, the coefficient of x is 8 and the coefficient of x^2 is 30

(b) Find the value of n and the value of a

(4)

(c) Calculate the coefficient of x^3 in $f(x)$

(2)

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6. Question 5 · part a · 2 marks · Working and answer

Working and answer

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7. Question 5 · part b · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

5

$$f(x) = (1 + ax)^n \text{ where } n > 2$$

(a) Write down the expansion of $f(x)$ in ascending powers of x up to and including the term in x^3
Give each coefficient in terms of n and a

(2)

In the expansion of $f(x)$, the coefficient of x is 8 and the coefficient of x^2 is 30

(b) Find the value of n and the value of a

(4)

(c) Calculate the coefficient of x^3 in $f(x)$

(2)

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7. Question 5 · part b · 4 marks · Working and answer

Working and answer

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8. Question 5 · part c · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

5

$$f(x) = (1 + ax)^n \text{ where } n > 2$$

(a) Write down the expansion of $f(x)$ in ascending powers of x up to and including the term in x^3
Give each coefficient in terms of n and a

(2)

In the expansion of $f(x)$, the coefficient of x is 8 and the coefficient of x^2 is 30

(b) Find the value of n and the value of a

(4)

(c) Calculate the coefficient of x^3 in $f(x)$

(2)

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8. Question 5 · part c · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 8 marks)

8. Question 5 · part c · 2 marks · Working and answer

Working and answer

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9. Question 6 · part b · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.

6 (a) Show that

$$\frac{a}{\sqrt{4+bx}} = \frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$$ where a and b are positive integers.

(2)

The expansion of $\frac{a}{\sqrt{4+bx}}$ in ascending powers of x can be written as

$$P+Qx+Rx^2+Sx^3$$

where P, Q, R and S are rational numbers.

(b) Show that $Q=-\frac{ab}{16}$ and $S=-\frac{5ab^3}{2048}$

and find P and R in terms of a and b, as fractions in their lowest terms.

(4)

Given that $Q=\frac{128}{5}S$ and $R=\frac{9}{256}$

(c) show that $a=3$ and $b=1$

(3)

(d) Hence, using an appropriate value of x, find, to 3 decimal places, an approximate value for $\frac{\sqrt{6}}{2}$

(3)

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9. Question 6 · part b · 4 marks · Working and answer

Working and answer

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10. Question 6 · part c · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

6 (a) Show that

$$\frac{a}{\sqrt{4+bx}} = \frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$$ where a and b are positive integers.

(2)

The expansion of $\frac{a}{\sqrt{4+bx}}$ in ascending powers of x can be written as

$$P+Qx+Rx^2+Sx^3$$

where P, Q, R and S are rational numbers.

(b) Show that $Q=-\frac{ab}{16}$ and $S=-\frac{5ab^3}{2048}$

and find P and R in terms of a and b, as fractions in their lowest terms.

(4)

Given that $Q=\frac{128}{5}S$ and $R=\frac{9}{256}$

(c) show that $a=3$ and $b=1$

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(d) Hence, using an appropriate value of x, find, to 3 decimal places, an approximate value for $\frac{\sqrt{6}}{2}$

(3)

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10. Question 6 · part c · 3 marks · Working and answer

Working and answer

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11. Question 3 · part b · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

3 (a) Expand

$$(1 + px)^{-5} \quad p \neq 0$$

in ascending powers of x, up to and including the term in x^4
Give each term in its simplest form.

(3)

The coefficient of x^r in the expansion is c_r

Given that $c_4 = 2c_3$

(b) find the value of p.

(2)

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11. Question 3 · part b · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 5 marks)

11. Question 3 · part b · 2 marks · Working and answer

Working and answer

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12. Question 9 · part d · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

9 (a) Expand $(1 + 2x)^{-\frac{1}{3}}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as a fraction in its lowest terms.

(3)

(b) Find the range of values of x for which your expansion is valid.

(1)

$$f(x) = \frac{2 + kx^2}{(1 + 2x)^{\frac{1}{3}}}$$

(c) Obtain a series expansion of f (x) in ascending powers of x up to and including the term in x^3
Give your coefficients in terms of k where appropriate.

(3)

The coefficient of x^3 in the series expansion of f (x) is $-\frac{8}{3}$

(d) Find the exact value of k

(2)

(e) Hence, using algebraic integration, estimate the value of

$$\int_{0.1}^{0.2} f(x)\,dx$$

Give your answer to 4 decimal places.

(5)

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12. Question 9 · part d · 2 marks · Working and answer

Working and answer

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13. Question 7 · part c · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.

7 (a) Expand $\left(1+2x^2\right)^{-\frac{3}{4}}$ in ascending powers of x up to and including the term in x^6

Express each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{(2+kx)}{\left(1+2x^2\right)^{\frac{3}{4}}} \quad \text{where } k \neq 0$$

(b) Obtain a series expansion for $f(x)$ in ascending powers of x up to and including the term in x^5

Give each coefficient in terms of k where appropriate.

(2)

The coefficient of the term in x^5 is fourteen times the coefficient of the term in x^2

(c) Find the value of k

(2)

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13. Question 7 · part c · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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13. Question 7 · part c · 2 marks

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Further practice · Attempt part c only.

Question 7 continued

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13. Question 7 · part c · 2 marks

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Question 7 continued

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(Total for Question 7 is 7 marks)

13. Question 7 · part c · 2 marks · Working and answer

Working and answer

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